

What is a True Semantic Layer?

Building Enterprise Knowledge Architectures for AI Readiness

Comprehensive Guide with Visual Perspectives
Generated: October 08, 2026

1. Introduction: The AI Methods Landscape

Artificial Intelligence is not a monolithic discipline. Rather, it represents a panoply of complementary and overlapping methodologies, each addressing different aspects of intelligent systems.



No. 1: Comprehensive, heterogeneous data

- All sorts of data can be meaningfully brought together, not just tabular data.
- Described documents and other media can articulate the context that tables are a part of.
- Different forms of data and knowledge brought together reinforce and support each other, boosting overall reusability.



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**COMPREHENSIVE,
HETEROGENEOUS DATA**

AI Methods Landscape

The Layered Nature of AI: AI encompasses symbolic AI (rule-based), machine learning (statistical), deep learning (neural networks), generative AI (content creation), and NLP (language understanding). Importantly, symbolic AI methods—historically dominant in enterprise IT—are experiencing a renaissance, now supercharged by modern tooling.

2. The Funding Gap: Where Investment Falls Short

Most funding in the AI industry is concentrated in a few high-profile companies. Databricks, OpenAI, and others have captured the lion's share of venture capital. However, this concentration masks a critical gap: **most funding sources are not focused on solving the biggest data problems.**



No. 5: Means for unambiguous, AI-ready data





LINKED, RELATIONSHIP-RICH DATA



DEDICATED HUMANS IN THE LOOP



RATIONAL, SHARED CONTEXT



SHARED, COMPOSABLE METADATA MODELS

"We need the AI to know, not guess, that a specific pump (P-101) is connected to a motor (M-101), requires a specific lubricant (ISO VG 460), and has a maintenance schedule tied to runtime hours. A shared ontology – a knowledge 'dictionary' – becomes essential, ensuring every term has one unambiguous meaning, traceable across multiple languages."

"The AI community often refers to this structured, interconnected knowledge base as a 'knowledge graph.' Every table becomes a set of complete statements, every diagram – a structured text file, every chart – its description."
–Yuriy Yuzifovich of Hitachi, "Industrial AI — Where knowledge (management) is power," *CIO*, Oct 6, 2025

Funding Gap Analysis

Real Problems: Data quality & governance, knowledge lifecycle management, semantic interoperability, and context/meaning. Companies like Databricks are data-oriented but perpetuate the status quo. They haven't addressed the root problems of data and knowledge lifecycle management—this is where semantic layers become essential.

3. Avoiding Toxic Software: Standards & Humanistic Design

Enterprise software systems accumulate complexity, fragmentation, duplication, and bloat over time. To avoid this, organizations must embrace open standards and humanistic architectural principles.

No. 3: FAIR data architecture



Data systems by design are relationship-rich, standards-based and intended to encourage sharing and reuse.

Principle	Objective
Findable	Discovery: Making sure humans and computers can locate the data.
Accessible	Retrieval: Defining how a user or system can get to the data once found.
Interoperable	Integration: Ensuring data can be used with other datasets and tools.
Reusable	Optimized Use: Ensuring the data can be used by others long-term.

Distilled from <https://www.go-fair.org/fair-principles/>



Toxic Software Standards

Key Principles: Open standards (RDF, OWL, FAIR), humanistic architecture (human understanding first), simplicity, visibility, decentralization, and human in the loop. Recommended books: "Software Architecture: The Data-Centric Revolution" and forthcoming "Data-Centric Approach" provide frameworks for building architectures that resist toxicity.

4. Understanding Semantic Layers: Architecture

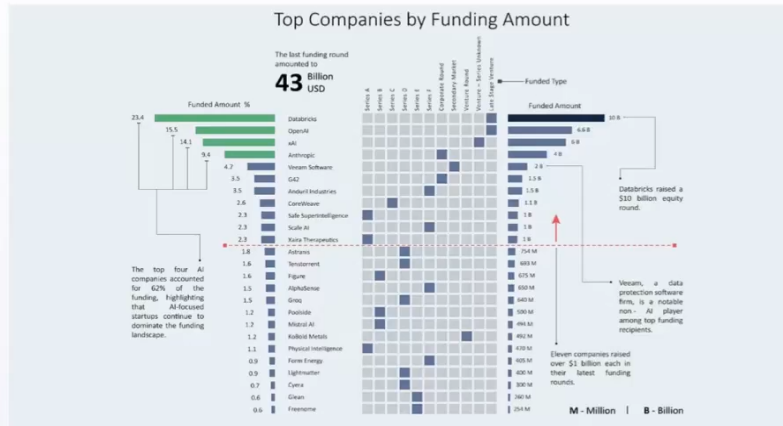
A semantic layer is a unified, machine-readable knowledge architecture that sits between raw data and business applications. It connects raw data with business users by offering a unified, easy-to-understand view.

Most funding sources aren't focused on helping solve the biggest data problems

Databricks and Scale AI are data-oriented, but perpetuate the status quo....

Companies like these have not been helping to disambiguate data outside their narrow focus.

They aren't fixing
the root problems
with data and
knowledge
lifecycle
management.



Brian Buntz, "25 landmark R&D-heavy tech funding rounds of 2024," at <https://www.rdworldonline.com/25-landmark-rd-heavy-tech-funding-rounds-of-2024/024>, December 18, 2024.

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Example Semantic Layer

Key Components: A semantic layer provides unified data abstraction, encodes business logic, maintains governed metadata, ensures interoperability (standards-based), and enables AI readiness through explicit semantics. It is NOT simply a data transformation layer—it is a fundamental architectural component that encodes meaning itself.

5. Real-World Implementation: EY's Knowledge Hub

Ernst & Young built a firmwide semantic knowledge hub demonstrating how a true semantic layer functions in complex, multi-system enterprise environments.



Help companies steer clear of toxic software

Back open standards and humanistic architecture

IT's biggest problems

- Overwhelming complexity
- Fragmentation
- Duplication
- Bloat
- Anti-human AI initiatives

Emancipating goals and architecture

- FAIR data and knowledge sharing
- Visibility across supply chains
- Decentralized web, storage and identity
 - SOLID
 - DIDs
- Human in the loop
- Self-reliance
- Ease of use

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Tell-all books that have helped companies shift direction



To be released in November:
Data-Centric Accounting

A tell-all speaker at this event:
Michael Iantosca of Avalara

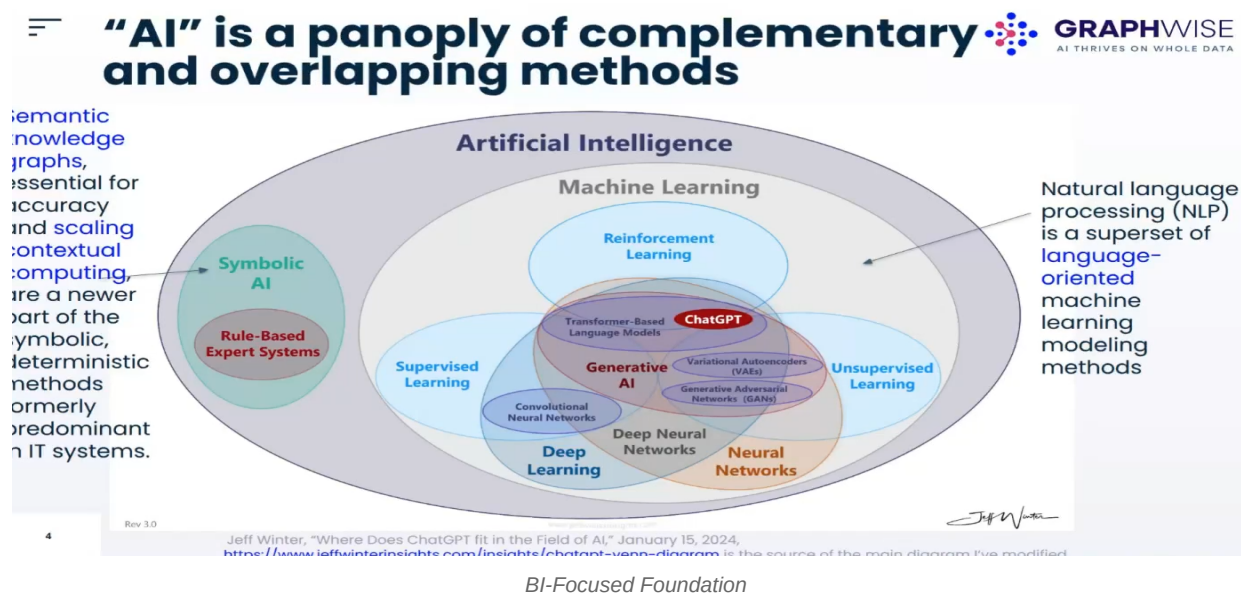
A radical standard worth checking out:
Zero-Copy Integration
(CA Digital Governance Council)

EY Knowledge Hub

Key Achievement: EY's semantic backbone reduces misclassification errors and increases content uptake by up to 60%. The hub includes GraphRAG applications, enterprise knowledge graphs, domain knowledge models, consistent semantic metadata, and multimodal data integration. This demonstrates the transformative power of semantic clarity in enabling faster, more accurate retrieval and reasoning.

6. Beyond BI: Knowledge as Foundation, Not Data

Business Intelligence tools focus on **data**—tables, metrics, dashboards. This data-centric approach has dominated enterprise architecture for decades. However, BI foundations are missing a critical layer: **knowledge**.



The Gap: BI tools support data access but not knowledge sharing. Data can be correct without its semantics being correct. Sharing knowledge requires explicit context and semantics. GenAI needs semantic grounding—context and meaning enabling accurate reasoning. As Malcolm Hawker notes: *"The history of everything we do in business could certainly be captured in an ongoing story."* This 'story'—the narrative context and semantic grounding—is what transforms raw data into actionable knowledge.

7. Semantics is Context, Not Just 'Correctness'

Data quality today is not just correctness of the data. It's also about the semantics being correct—what to represent in that data.

Semantics is fundamentally about context. The word 'Paris' could refer to the city, a hotel chain, or a person. The meaning depends entirely on context. The shift from keyword search to semantic/concept search illustrates this transformation. In keyword search, users hope the system returns relevant results. In semantic search, the system understands the entities and relationships, returning results based on conceptual understanding.

As Sumit Pal notes: "Semantics is often hidden in the data. When humans hear 'Paris', they think about the city, not Paris Hilton. And when they hear 'Hilton', they think about the hotel chain. So it's all about context."

8. Five Pillars of a True Semantic Layer

A true semantic layer for enterprise AI readiness rests on five critical pillars:

1. Comprehensive, Heterogeneous Data:

All sorts of data—tables, documents, media—can be meaningfully brought together. Described documents articulate context that tables alone cannot.

2. Knowledge-Centric Culture:

Knowledge and data must consistently benefit business needs throughout their lifecycles—from collection to archiving.

3. FAIR Data Architecture:

Findable, Accessible, Interoperable, Reusable. Data systems designed by FAIR principles encourage sharing and reuse.

4. Advanced Identity & Ownership:

First-person identity combined with cohesive global identity schemes ensures each entity has a resolvable, context-rich identifier.

5. Unambiguous, AI-Ready Data:

Linked, relationship-rich data with rational, shared context. Every term has one unambiguous meaning, traceable across languages and systems.

9. Getting Started: Implementation Strategy

Building a semantic layer is a long-term commitment, but it needn't be overwhelming. The most successful approaches start small, demonstrate value early, and iterate systematically.

1. Ground Your Enterprise in a Unifying Vocabulary/Ontology:

Rather than building one massive ontology, start with a focused core representing your most critical business concepts.

2. Show Value Each Quarter:

Build a small, demonstrable piece of your vision now. Quick wins build momentum and support.

3. Embrace the TOAST Method:

Tiny Ontologies All Strong Together—build small, focused ontologies for specific domains, then compose them.

4. Leverage AI-Assisted Modeling:

Tools like Graphwise can bootstrap ontologies in minutes, not months. Use AI to accelerate modeling, then validate with domain experts.

The Bottom Line: A true semantic layer is not a one-time project; it's an ongoing practice of making meaning explicit, shareable, and machine-actionable. Organizations that invest in semantic clarity will lead in trustworthy, explainable, and cost-effective AI delivery.